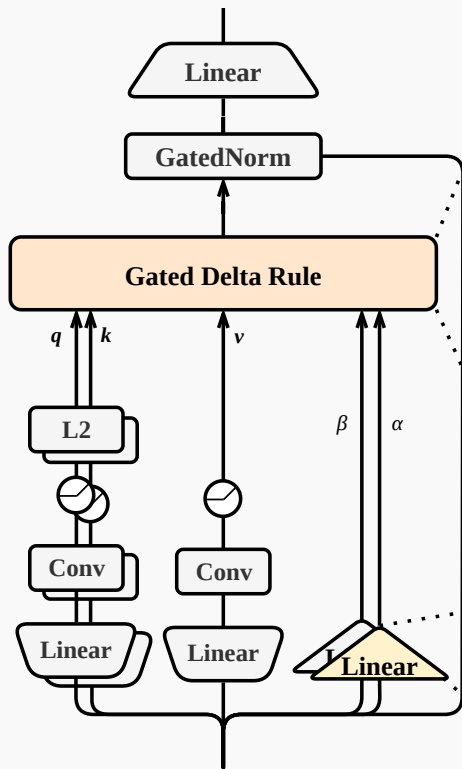
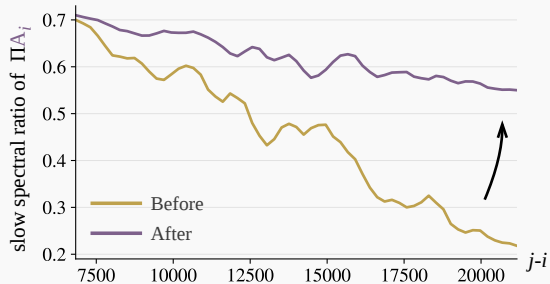


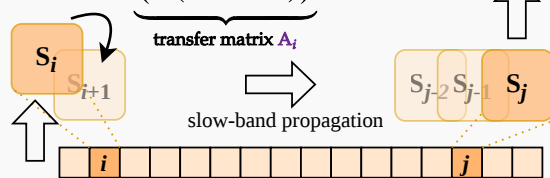
Gated DeltaNet



Understanding Long-Context Retrieval through Spectral Dynamics



$$\mathbf{S}_{i+1} = \left(\alpha_i (\mathbf{I} - \beta_i \mathbf{k}_i \mathbf{k}_i^T) \right) \mathbf{S}_i + \beta_i \mathbf{v}_i \mathbf{k}_i^T$$



SpectralShift for Long-Context Continual Pretraining

$$\mathbf{W}_\alpha, \eta^{\text{tar}}$$

Alpha Proj.



$$\mu \mathbf{J} + s \cdot (\mathbf{W}_\alpha - \mu \mathbf{J}), s \cdot \eta^{\text{tar}}$$

Alpha Proj.